

Exercise 1.

```
• (pytorch-env) (base) mayageva@197 pytorchtest % python  
PyTorchTest.py  
Pytorch will use:  cpu  
<class 'torch.Tensor'>
```

Exercise 2.

```
epoch68 validation loss = tensor(5.0533e-05)  
epoch69 validation loss = tensor(4.9131e-05)  
epoch70 validation loss = tensor(4.7737e-05)  
epoch71 validation loss = tensor(4.6401e-05)  
epoch72 validation loss = tensor(4.5093e-05)  
epoch73 validation loss = tensor(4.3831e-05)  
epoch74 validation loss = tensor(4.2614e-05)  
epoch75 validation loss = tensor(4.1421e-05)  
epoch76 validation loss = tensor(4.0255e-05)  
epoch77 validation loss = tensor(3.9130e-05)  
epoch78 validation loss = tensor(3.8021e-05)  
epoch79 validation loss = tensor(3.6953e-05)  
epoch80 validation loss = tensor(3.5916e-05)  
epoch81 validation loss = tensor(3.4910e-05)  
epoch82 validation loss = tensor(3.3934e-05)  
epoch83 validation loss = tensor(3.2972e-05)  
epoch84 validation loss = tensor(3.2055e-05)  
epoch85 validation loss = tensor(3.1150e-05)  
epoch86 validation loss = tensor(3.0281e-05)  
epoch87 validation loss = tensor(2.9431e-05)  
epoch88 validation loss = tensor(2.8601e-05)  
epoch89 validation loss = tensor(2.7788e-05)  
epoch90 validation loss = tensor(2.7010e-05)  
epoch91 validation loss = tensor(2.6264e-05)  
epoch92 validation loss = tensor(2.5527e-05)  
epoch93 validation loss = tensor(2.4815e-05)  
epoch94 validation loss = tensor(2.4112e-05)  
epoch95 validation loss = tensor(2.3426e-05)  
epoch96 validation loss = tensor(2.2777e-05)  
epoch97 validation loss = tensor(2.2136e-05)
```

Exercise 2 Question A: How many epochs is the model trained for?
1000 for the first loop, and 100 for the second.

Exercise 2 Question B: Is the loss function in this code mean square error or cross entropy?
The loss function is mean square error

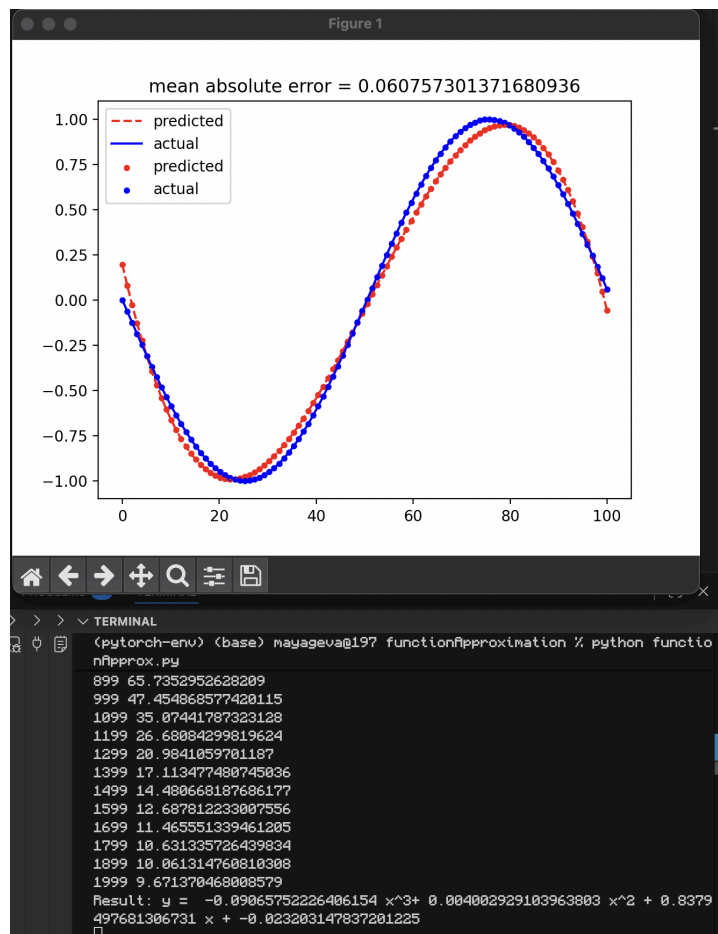
Exercise 2 Question C: Why is it important to evaluate the validation loss during training?
Evaluating the validation loss is important because it can detect overfitting or underfitting. With this information we can assess whether continuing to train the model will benefit or harm the accuracy.

Exercise 2 Question D: How many trainable parameters does the simple model have?
2, weight and bias

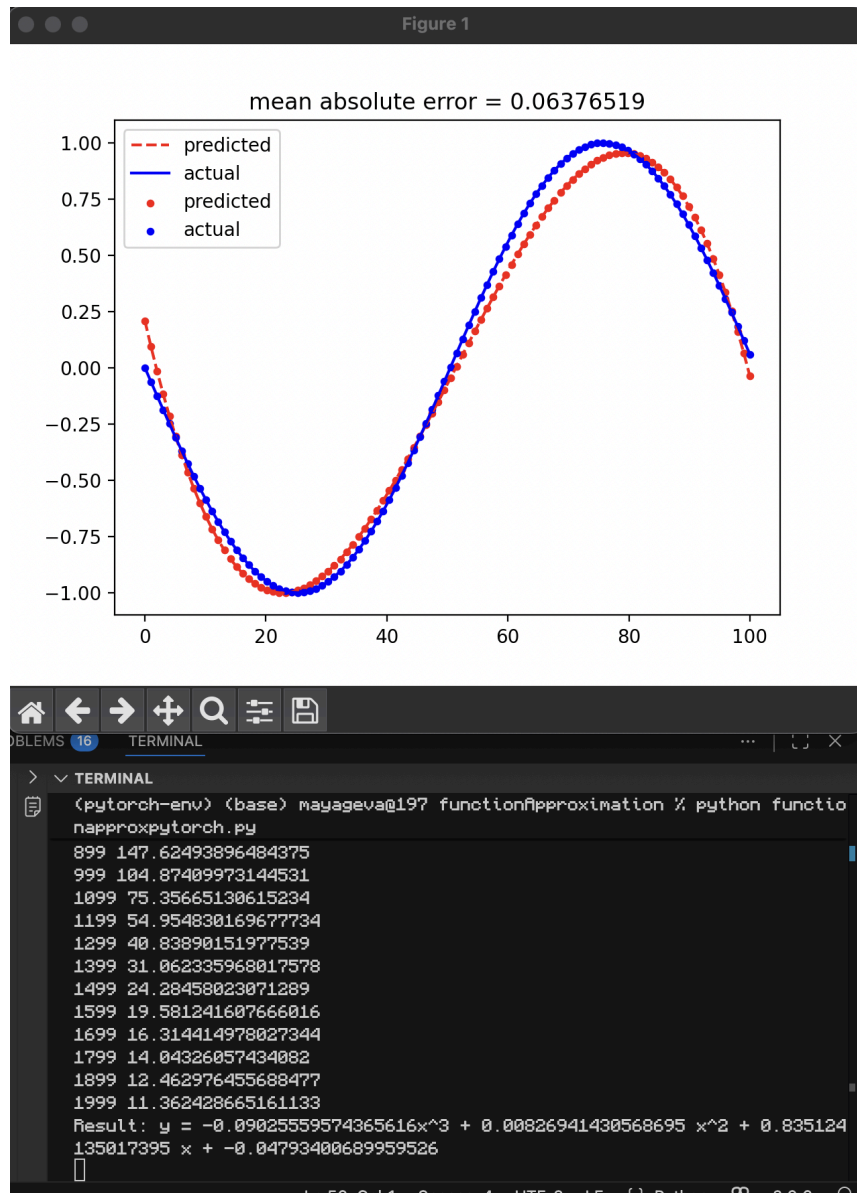
Exercise 2 Question E: In this training algorithm is full batch, mini-batch or stochastic gradient descent being used? Hint: This does not depend on the optimizer in the code but rather the batch size of the dataloader.

The batch size of the dataloader is 4, which is greater than 1 but smaller than the whole data set. This means the model is using mini-batches to train.

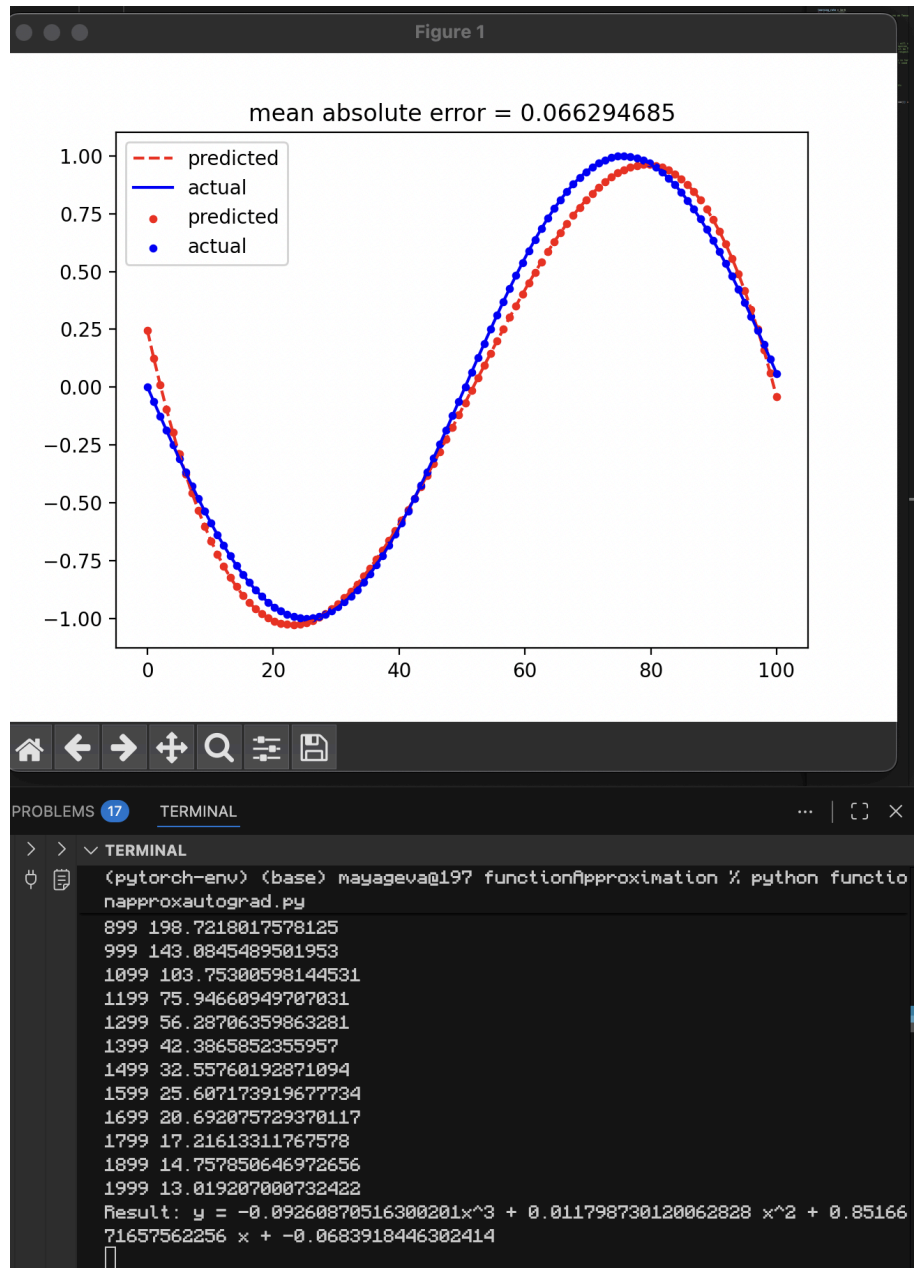
Exercise 3:



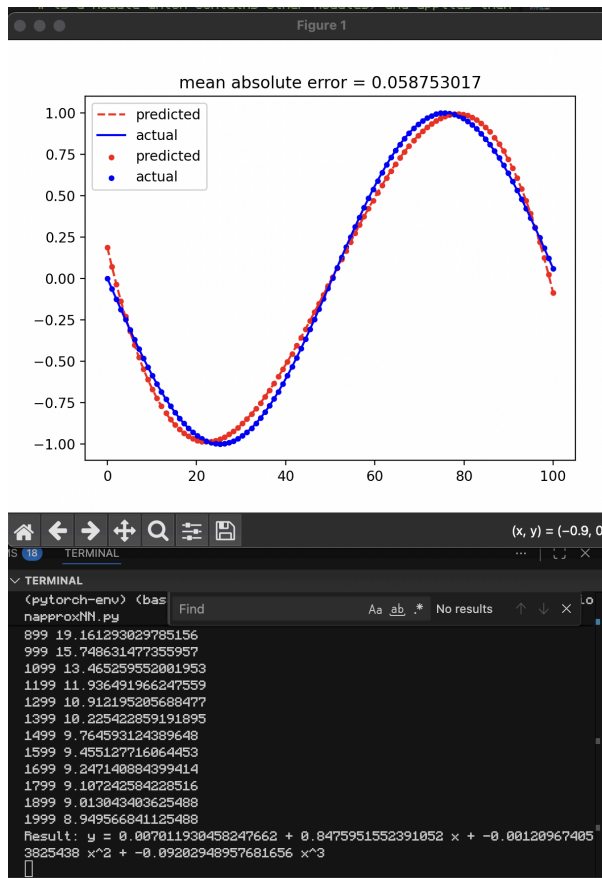
Exercise 4:



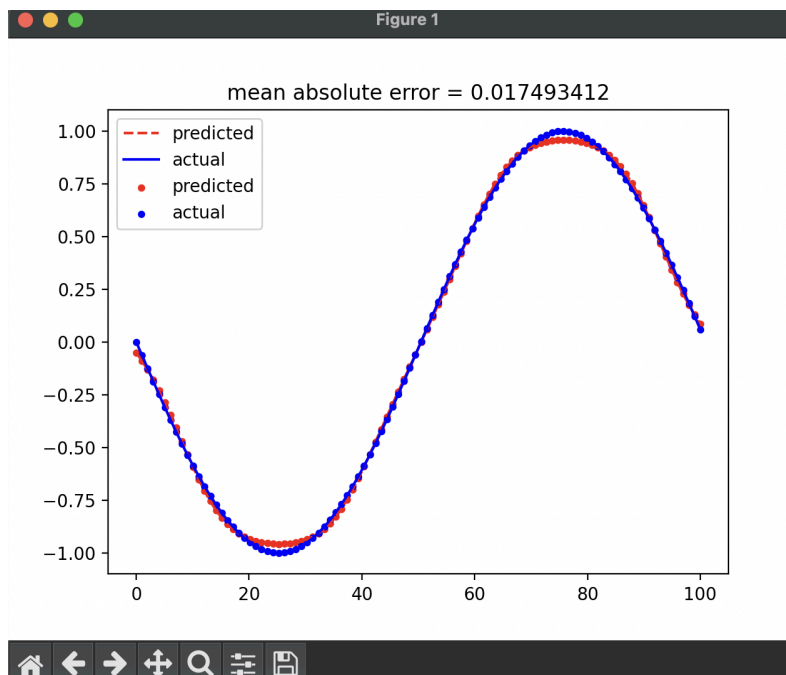
Exercise 5:



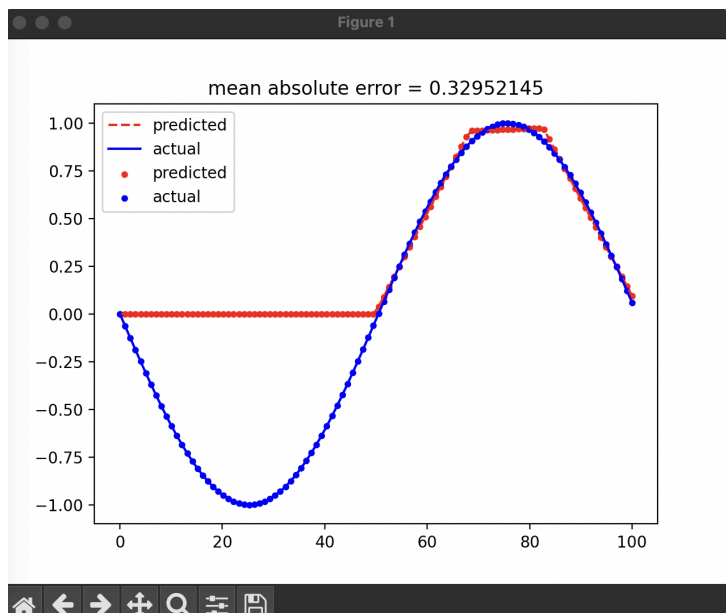
Exercise 6:



Exercise 7:

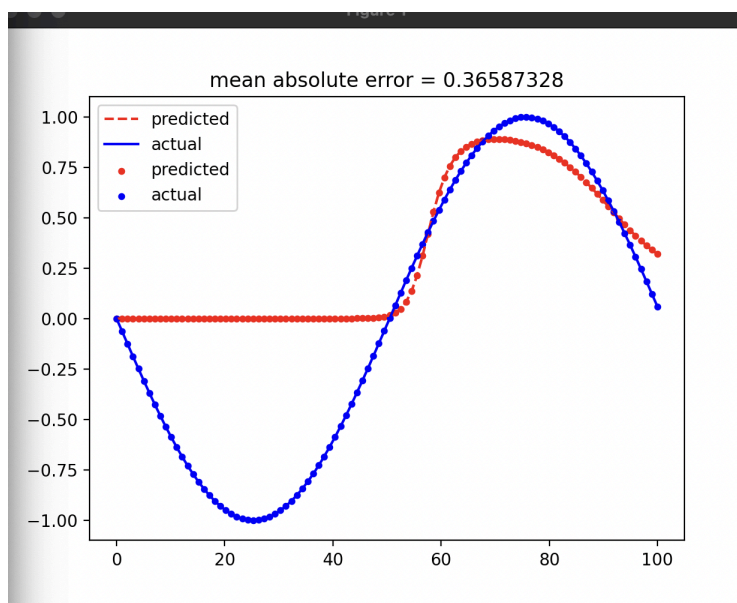


Exercise 8 A:



$X=0$ and $X=40$ are supposed to have negative outputs (as seen when we used `torch.nn.TanH()`), but ReLu cannot learn negative values, and can only give positive outputs, so ReLu defaulted them to 0, which is incorrect

Exercise 8 B:



Sigmoid has the same issue with not being able to produce negative values. Sigmoid can produce values from 0 to 1, while the actual sine function goes from -1 to 1, so the negative values are never learned.

Exercise 9:

```
super().__init__(loader)
torch.Size([3, 224, 224])
epoch: 0 186 1.5928934264183043
epoch: 1 186 1.457607918381691
epoch: 2 186 1.3941665416955948
epoch: 3 186 1.344834640622139
epoch: 4 186 1.3090266644954682
epoch: 5 186 1.277155522108078
epoch: 6 186 1.2670344734191894
epoch: 7 186 1.235311776995659
epoch: 8 186 1.2193710219860077
epoch: 9 186 1.1989673459529877
epoch: 10 186 1.1913249856233596
epoch: 11 186 1.1839808112382888
epoch: 12 186 1.1714291602373124
epoch: 13 186 1.1628590589761734
epoch: 14 186 1.1569496726989745
epoch: 15 186 1.1489774215221404
epoch: 16 186 1.1437997114658356
epoch: 17 186 1.1396750319004059
epoch: 18 186 1.136100670695305
epoch: 19 186 1.131957421898842
epoch: 20 186 1.1298451137542724
epoch: 21 186 1.127543767094612
epoch: 22 186 1.1260214251279832
epoch: 23 186 1.1239161384105683
epoch: 24 186 1.1230291324853896
accuracy: 0.7446428571428572
```